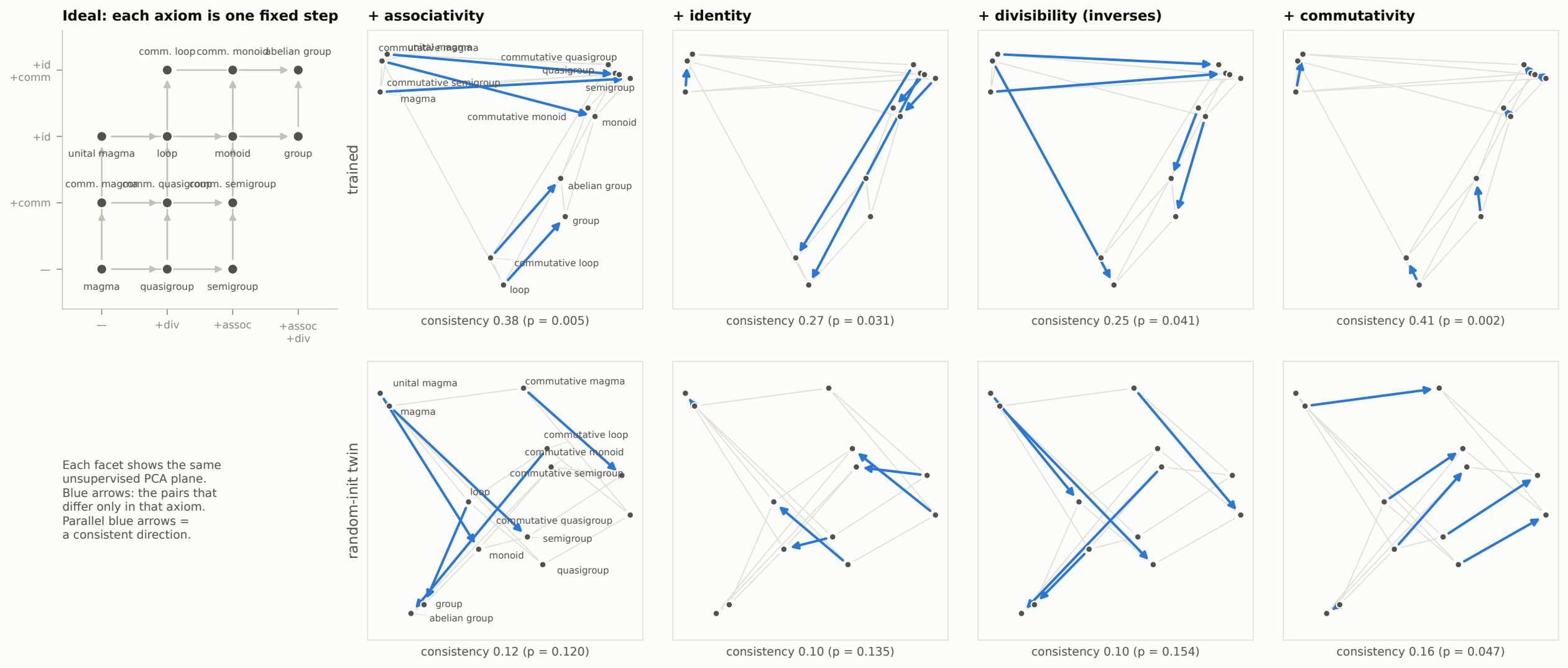
Ideal: each axiom is one fixed step
+ associativity
+ identity
+ divisibility (inverses)
+ commutativity
trained
random-init twin
+id +comm
+id
+comm
—
comm. loop
comm. monoid
abelian group
unital magma
loop
monoid
group
comm. magma
comm. quasigroup
comm. semigroup
magma
quasigroup
semigroup
—
+div
+assoc
+assoc +div
commutative quasigroup
quasigroup
commutative semigroup
semigroup
commutative monoid
monoid
abelian group
group
commutative loop
loop
commutative magma
consistency 0.38 (p = 0.005)
consistency 0.27 (p = 0.031)
consistency 0.25 (p = 0.041)
consistency 0.41 (p = 0.002)
consistency 0.12 (p = 0.120)
consistency 0.10 (p = 0.135)
consistency 0.10 (p = 0.154)
consistency 0.16 (p = 0.047)
Each facet shows the same unsupervised PCA plane. Blue arrows: the pairs that differ only in that axiom. Parallel blue arrows = a consistent direction.